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Subject: OS Lab	**Sem: III**	**AY: 2026-27**
EXPERIMENT No : 1		**Assignment No: 1**
TITLE : Introduction to Unix and Linux Fedora Operating System.		**DoP : 07-07-2026**

Aim : Case Study assignment on operating systems (Windows, Linux, Solarise, MSDOS, Macintosh, iOS based, Android Based , other, Select any four)

Learning Outcome: *1. To understand the features of existing OS*

2. To Compare the OS based on features, overheads, and applications

Theory:

1. ReactOS

- Introduction

ReactOS is an OS that is open-source and free and is made to work well with Microsoft Windows applications and drivers. It enables people to run Microsoft applications and does not require any license to use. ReactOS is easy to use, lightweight and it is aimed at users who would like an alternative to Windows at no cost.

It is developed from scratch and it is not derived from Linux and Windows OSes. Its primary objective is to make an operating system that resembles Windows OS yet being open-source. Though ReactOS is in its development process, it is widely used by developers, students and tech enthusiasts interested in operating systems.

ReactOS is compatible with many Windows applications and hardware drivers. Its primary application includes testing and developing software and learning about operating systems.

- System Requirements

The basic requirements to install ReactOS are:

* Pentium-compatible processor or higher
* 512 MB RAM (1 GB recommended)
* 2 GB free storage space

These requirements make ReactOS suitable for older computers with limited hardware resources.

- Advantages

ReactOS provides several benefits to users.

* Completely free and open source
* Lightweight and fast
* Compatible with many Windows applications
* Supports several Windows hardware drivers
* Can run on older computers
* No license cost

- Disadvantages

ReactOS also has some limitations.

* Still under development

* Some Windows applications may not work properly

* Less stable than commercial operating systems

* Limited hardware compatibility

- Applications (Where It Is Used)

ReactOS is used in several areas, including:

* Educational institutions for learning operating systems

* Software testing and application compatibility testing

* Programming and operating system research

* Open-source software development

* Computer laboratories

2. FreeBSD Operating System

- Introduction

FreeBSD is an open source operating system derived from the Berkeley Software Distribution (BSD). BSD itself is a version of the Unix operating system, and it is characterized by its outstanding performance, stability, and security. FreeBSD is commonly used in server computers, network systems, cloud computing services, and embedded systems.

Contrary to many other operating systems, the FreeBSD operating system comprises both the operating system software itself and all the system utilities. Thus, it is very stable and easy to administer. Also, it runs on multiple hardware architectures and provides advanced networking capabilities. Thanks to its outstanding performance, FreeBSD is preferred by various corporations.

The development of FreeBSD is undertaken by a big community of developers, who update it, fix bugs, and implement new functionalities on a regular basis. FreeBSD is available absolutely free of charge.

- System Requirements

The minimum requirements to install FreeBSD are:

* 64-bit processor (Intel or AMD)
* 2 GB RAM (4 GB recommended)
* 20 GB free hard disk space

- Advantages

FreeBSD provides many benefits to users and organizations.

* Free and open-source software
* Very stable and reliable
* Strong security features
* Excellent networking performance
* Supports multiple hardware platforms

- Disadvantages

Although FreeBSD has many strengths, it also has some limitations.

* Not as beginner-friendly as Windows
* Smaller desktop software collection
* Some hardware devices may have limited driver support
* Requires knowledge of Unix commands

- Applications (Where It Is Used)

FreeBSD is used in many industries because of its speed and reliability.

* Web servers and hosting companies
* Cloud computing platforms
* Internet service providers (ISPs)
* Network security systems
* Educational institutions
* Research laboratories
* Storage servers
* Firewall and router systems
* Data centers
* Software development and testing

3. Haiku Operating System

- Introduction

Haiku is an operating system that is free and open-source. The idea behind Haiku is a rapid operation, a straightforward installation process, and convenient usage. This OS is similar to BeOS and is built for speed, fast start-up time, and user-friendly desktop environment. Haiku is lightweight and can work on any computer, including both old and new models.

It is worth mentioning that the operating system is aimed at private users and their needs. The system is characterized by a clear graphical interface, fast file management, and multitasking capabilities. Being an open-source system, it attracts many developers all over the world. It is widely known among students, programmers, and other people interested in operating systems.

Despite the fact that Haiku is not as popular as Windows or Linux, it is valued for its speed, convenience, and modernity. It is a great option for everyone willing to get acquainted with another operating system.

- System Requirements

The basic requirements for installing Haiku are:

* x86 or x86-64 processor
* 2 GB RAM (minimum)
* 8 GB free hard disk space

- Advantages

Haiku offers several benefits that make it unique.

* Free and open-source operating system
* Very lightweight and fast
* Quick boot time
* Easy-to-use graphical interface
* Low hardware requirements
* Smooth multitasking
* Stable performance for everyday tasks
* Ideal for learning operating system concepts

- Disadvantages

* Smaller software library than Windows or Linux
* Limited hardware driver support
* Not suitable for heavy gaming
* Some modern applications are unavailable
* Smaller user and developer community
* Still under active development

- Applications (Where It Is Used)

Haiku is used in several areas, including:

* Educational institutions
* Computer science laboratories
* Operating system research
* Software development
* Open-source community projects
* Personal computers for basic tasks
* Learning programming and system design
* Technology enthusiasts and hobbyists

4. ChromeOS Operating System

- Introduction

ChromeOS is an operating system created by Google. The main purpose of this operating system is to be used for web computing and cloud services. In contrast to other operating systems, ChromeOS is used mostly to run applications through Google Chrome web browser and Android applications. This operating system is very light, secure and user-friendly, which makes it quite popular among students, teachers, and office workers.

Mainly ChromeOS is installed on laptops that are named Chromebooks. One of the main characteristics of these devices is that they have very quick start-up, they have great battery life and they perform very smoothly. Due to the fact that most of the files and applications are kept in the cloud, people can access their information from any device by using their Google account.

ChromeOS updates itself automatically in the background.

- System Requirements

The basic requirements for ChromeOS are:

* Chromebook or ChromeOS-supported device
* Intel or ARM processor
* Minimum 4 GB RAM
* 32 GB storage or higher

- Advantages

ChromeOS provides many useful features.

* Fast boot time
* Easy and simple interface
* Strong security against viruses and malware
* Automatic software updates
* Long battery life
* Supports Android and Linux applications
* Cloud storage integration with Google Drive
* Low maintenance and easy management

- Disadvantages

ChromeOS also has some limitations.

* Requires an internet connection for many features
* Limited support for traditional Windows software
* Less suitable for high-end gaming

* Professional editing software may not be available

* Most devices have limited storage capacity

- Applications (Where It Is Used)

ChromeOS is used in many places, such as:

* Schools and colleges

* Online education

* Offices and businesses

* Government organizations

* Libraries

* Call centers

* Home users for internet browsing

* Students for assignments and online learning

5. Solaris Operating System

- Introduction

It is a robust Unix operating system that was created for use in powerful computer machines and servers. Solaris OS has proved to be very stable and reliable in handling big amounts of work and also secure from various threats. The operating system is widely used in business enterprises where computers should operate continuously without stopping.

The software includes such features as multitasking, virtual memory management, file system protection, and strong networking capabilities. Moreover, Solaris OS is made to operate efficiently in both small and large computer systems. Also, it can support more than one user simultaneously.

The operating system has been used in many banks, government offices, telecommunication firms, and data centers due to its reliability and efficiency. Solaris is also recognized to be the most secure operating system for the server application.

- System Requirements

The basic requirements for Solaris are:

* 64-bit processor
* Minimum 2 GB RAM (4 GB recommended)
* At least 20 GB free hard disk space

- Advantages

Solaris provides many benefits for organizations.

* High system stability
* Excellent security features
* Strong networking performance
* Supports multiple users
* Efficient memory management
* Reliable for enterprise applications
* Good virtualization support
* Handles heavy workloads efficiently

- Disadvantages

Solaris also has some limitations.

* Not free for all enterprise versions

* Requires experienced administrators

* Limited desktop software compared to Windows

* Smaller user community

* Less hardware compatibility than Linux

* Not suitable for gaming or personal entertainment

- Applications (Where It Is Used)

Solaris is widely used in:

* Banking systems

* Government organizations

* Telecommunication companies

* Cloud computing platforms

* Enterprise database servers

* Data centers

* Financial institutions

* Research organizations

* Large business networks

Conclusion:

ReactOS is an open source operating system developed to be compatible with Microsoft Windows. The primary intention of using ReactOS is that the user should run Windows programs and drivers without paying for the Windows operating system. It is small and helpful for students, developers, and those who wish to learn operating systems. However, even though it is not fully completed, ReactOS serves as a good example of an open source and Windows-compatible OS.

FreeBSD is a Unix-like operating system known for its reliability, security, and excellent performance. FreeBSD is popularly used in web servers, cloud computing, Internet services, and networking. Organizations use FreeBSD since it can operate for a long time without any issues.

Haiku is an efficient and light-weight operating system derived from the BeOS operating system. This is a user-friendly and speedy operating system which starts up quickly and performs well with minimal hardware. The operating system consumes less amount of memory and operates well even in old machines. It is used mainly for education, software testing, research, and learning purposes.

ChromeOS is produced by Google Inc. It is used mainly in Chromebook laptops. It is an internet-based and cloud-computing platform. Chrome OS has quick start-up, automated update process, strong security, and support for android apps. It is widely used in universities, offices, colleges, and schools by teachers and students because of its simplicity.

Solaris is a very efficient Unix-based operating system designed for enterprise computers. It is famous for its reliability and secure nature along with heavy duty handling. The Solaris is widely used in banks, government sectors, telecommunication organizations, cloud services, and large scale data centers.

References:

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